

Rishabh Shah

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EDUCATION

Rutgers University – New Brunswick

M.S. in Computer Science, GPA: 3.83/4.0

Expected May 2027

Graduate Student

University of Mumbai

B.Tech. in Artificial Intelligence & Data Science, CGPA: 7.76/10.0 (Indian scale)

Dec 2021 – Jun 2025

Mumbai, India

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, SQL, C

Frameworks & Tools: Django, Next.js, REST APIs, JWT Auth, Docker, Git, GitHub, GitHub Actions (CI/CD), Linux

Databases & Cloud: PostgreSQL, MySQL, MongoDB, AWS, Azure, Google Cloud Platform (GCP)

Testing: Pytest, Vitest, Playwright

AI/ML: TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLO

PROFESSIONAL EXPERIENCE

Aaditya Technologies

Software Engineering Intern

Onsite – Mumbai, India

Jan 2025 – Jul 2025

- Reduced API response times exceeding 400ms by restructuring MySQL schemas and rewriting critical SQL paths in Python and Next.js, cutting average latency by 22%
- Eliminated unauthorized cross-service requests across 4 backend services by building JWT-authenticated REST APIs with role-based access control, enforcing identity verification on every internal call
- Containerized 4 backend services with Docker and automated AWS release workflows, reducing manual per-release deployment setup by 65% for a 4-person backend team

Creative Line

Web Development Intern

Onsite – Mumbai, India

Jun 2024 – Sep 2024

- Resolved a coupling bottleneck across 3 features by refactoring 7 shared-logic components into isolated responsibilities, cutting per-file size by 70% and reducing review complexity
- Traced P95 response times above 3100ms to missing indexes and absent connection pooling by rebuilding 3 Next.js routes with both fixes, dropping load time to 420ms
- Closed SQL injection exposure in user-facing forms by switching to parameterized queries and adding input validation across authentication and CRUD endpoints, eliminating direct raw-input-to-database paths

TECHNICAL PROJECTS

Snap-Interview | AI Interview Practice Platform

snapinterview.app

Next.js, TypeScript, PostgreSQL, Prisma, Groq API, AWS S3, ElevenLabs, Better Auth

- Built an async interview pipeline coordinating microphone capture, LLM streaming, TTS synthesis, and S3 uploads, eliminating audio lag and race conditions
- Eliminated malformed AI output by designing a schema-enforced evaluation engine that scores responses from 0 to 100 with structured feedback across communication, technical depth, and clarity
- Hardened the upload and auth surface by scoping presigned S3 keys to per-user prefixes, adding CSRF protection and rate limits across 4 endpoint types, and automating quality gates with a CI pipeline covering lint, typecheck, unit, and e2e tests

CseGraph | Repository Context Engine for Coding Agents

github.com/RishiiShah/Csegraph

Python, SQLite, Tree-sitter, MCP, FTS5

- Built a repository indexing pipeline that parses 22 languages with Tree-sitter into a WAL-mode SQLite dependency graph, with incremental SHA256-based refresh and a lexical index over symbols, signatures, and docstrings
- Reduced agent token consumption by 96% on the engine's codebase (144 files, 1,509 symbols, 6,570 edges) by building FTS5 BM25 retrieval with exact-name boosting, hub-aware BFS traversal, and deterministic byte-capped context selection
- Exposed retrieval through a 6-tool MCP stdio server with session-aware filtering and confidence tiers for AST-proven versus inferred edges

Traffic Violation Detection & Automated Ticketing System

github.com/RishiiShah/Traffic

IEEE ICICIS 2025 | Scopus Index

- Automated the violation-to-notification pipeline with frame-by-frame detection, trigger-point violation logic, S3 evidence storage, and Twilio SMS alerts, completing processing within source-video duration plus 10 seconds
- Implemented an automated evaluation harness for YOLOv11x/v12x nano-to-XL variants, selecting YOLOv12x at 93.76% end-to-end accuracy with 97.56% vehicle and 93.26% helmet classification
- Replaced traditional OCR that failed on angled low-light plates by integrating Groq Llama-4 Scout as a vision-based fallback with regex post-validation, reaching 92.7% plate extraction accuracy